Meeting Summaries of April 16

Date: April 16
Time: 18:00 – 18:50 (PDT)
Attendees: Prof. Jan, Dr. Jabed, Dr. Yue and Feng
Topic: Progress report on **Chapter 2** and **Chapter 3**; review of chemical stress assessment, PCA methodology, hierarchical clustering, and planned comparisons; discussion of presentation structure.

Questions, Feedbacks and Suggestions

- Prof. Jan: **Remove PC2** from the PCA analysis to align with Jane's previous thesis work. Feng should also implement the **min-max transformation** step to match Jane's original methods before introducing any new approaches.

  > Key point for the next meeting
  >
  > Replicate Jane's results *first*. Only after the replication is confirmed should Feng present any new or extended methods.

- Prof. Jan: Replace original station codes in all tables with **standardized integrated codes** (DR/SCL) for consistency across datasets. Update figure captions accordingly to reflect the specific transformed data used in each analysis.
- Prof. Jan: For the cluster analysis, use **planned comparisons** at different hierarchical levels (Group A vs. Group B, then B_1 vs. B_2) rather than a single omnibus ANOVA. This approach better reveals the environmental variables that distinguish the groups and will likely require 3–4 additional ANOVA tables for different habitat features.
- Dr. Jabed: Present the **complete dataset** in all analyses by merging all dataset components before running any figures or statistics. Results based on partial data should not be treated as final.
- Dr. Jabed: Adopt a structured **presentation format** for every chapter meeting: problem statement → objectives → methodology → results → conclusions. Do not simply highlight sentences from published articles.
- Dr. Jabed: Add a **conclusions section** at the end of each chapter presentation to summarize the key findings, rather than just recapping the work done.
- Dr. Jabed: Update and share the **thesis schedule/timetable** with the committee. If timing goes off track, revise and redistribute it promptly.
- Dr. Jabed and Prof. Jan: Add an **algorithm block with pseudocode** explaining the permutation test methodology to the appendix of Chapter 2.

- Prof. Jan: Add a direct **comparison of RelMax, SumRel, and the hazard (HZD) metric** in Chapter 2 to clarify how the three contamination assessment approaches relate to one another.

Summary of Discussion

Thesis Progress Overview

The committee reviewed Feng's progress on Chapters 2 and 3. Dr. Jabed stressed that the first priority is to **merge all dataset** components and rerun all analyses on the complete dataset. The committee also agreed to schedule regular in-person meetings on Mondays, Wednesdays, and Fridays (10:00–12:00) in a larger lecture room (Old Main or House of Learning) to support intensive work during April.

Chemical Stress Assessment and PCA

Feng presented updates on the chemical stress assessment using PCA and RDA methods. Prof. Jan identified that the current implementation diverges from Jane's thesis in two ways: PC2 was retained instead of being removed, and raw PC values were used instead of the SumRel method with a min-max transformation. These reproductions should be retained **before** new work is introduced. The committee also requested that Feng add **a side-by-side comparison** of the SumRel, MaxRel, and HZD-based approaches to highlight how the scoring and standardization differences affect contamination rankings.

Hierarchical Cluster Analysis and Planned Comparisons

The committee reviewed Feng's Ward's hierarchical clustering results, which identified distinct site groups including a transitional assemblage. Prof. Jan recommended switching from a single ANOVA to a **planned comparisons** strategy: first compare Group A (least-polluted, green samples) against Group B (orange and blue samples combined), then subdivide Group B into B_1 and B_2 for a second-level comparison. This hierarchical approach provides more informative results by isolating the environmental variables that explain each split.

Invertebrate Assemblage Classifier (Chapter 3)

Feng's consensus clustering analysis identified three distinct clusters, including a transitional assemblage. Prof. Jan noted unfamiliarity with the consensus clustering approach and asked whether more current classification methods might be available. For the next meeting, Feng will prepare a **presentation** (not a full chapter draft) that covers the ordination method comparisons and demonstrates replication of Jane's classification results.

Presentation Skills and Scheduling

Dr. Jabed emphasised that Feng's presentation style needs to improve. Recommendations included attending a **CELT presentation skills workshop**, studying online resources on scientific presentation, and consistently applying the problem–method–results–conclusions structure. Feng committed to implementing these improvements starting with the next chapter meeting.

Action Items

- Merge all dataset components and rerun all analyses with the **complete dataset**; email updated figures (spatial display showing numbers over lakes and rivers) to the committee by **Friday night**.
- **Remove PC2** from the PCA analysis as per Prof. Jan's feedback.
- Implement the **min-max transformation** in the contamination assessment to match Jane's original methods.
- Update all tables to use **standardized integrated codes** (DR/SCL) instead of original station codes; map original codes to integrated codes throughout.
- Perform ANOVA using the **planned comparisons** approach at hierarchical levels (A vs. B, then B_1 vs. B_2).
- Add an **algorithm block** with pseudocode for the RDA-based least-stressed site searching work and its permutation test methodology in the Chapter 2 appendix.
- To add a **comparison of RelMax, SumRel, and HZD** metrics in Chapter 2.
- **Prepare replication of Jane's results** for Chapter 3 before presenting any new methods at the next meeting.
- Check the **CELT website** for presentation skills workshops and explore online presentation resources.
- Book a **large lecture room** (Old Main or House of Learning) for the Monday/Wednesday/Friday 10:00–12:00 meetings.
- Email the committee regarding **availability** for the Monday/Wednesday/Friday 10:00–12:00 meeting times.

Follow-Up

Next meeting time: April 23, 18:00–18:50 (PDT)

Supportive Materials

- A meeting recording has been restored on local drive.